

Model Evaluation Review Report

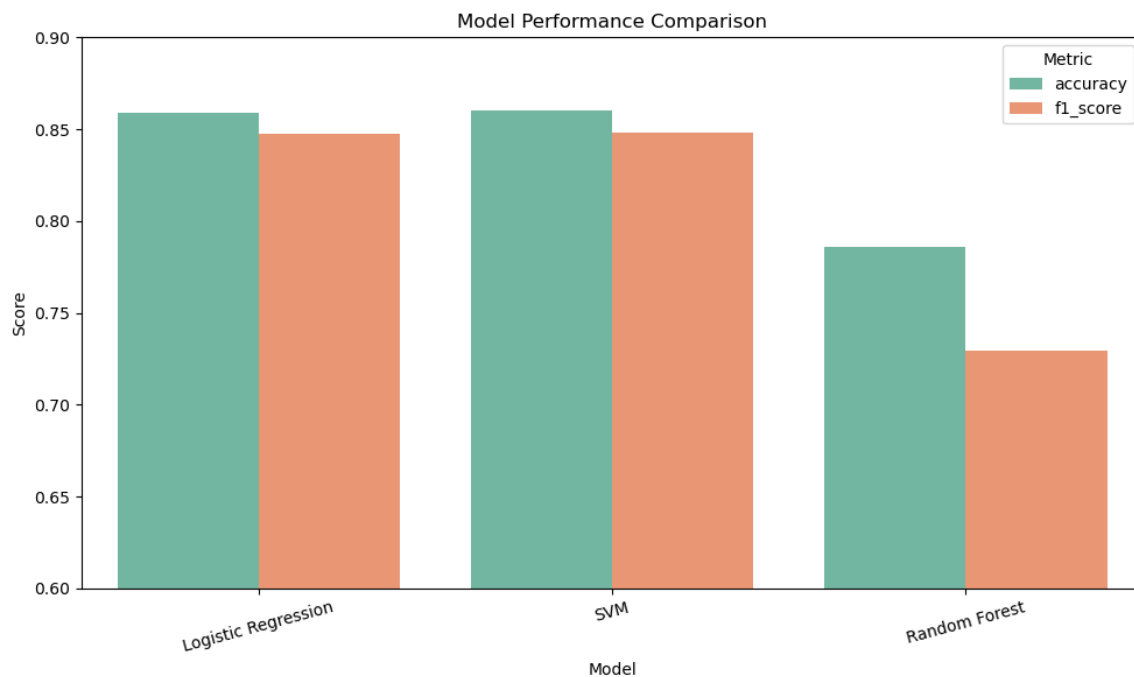
Dataset Summary

- **Total Samples:** 74,907 cleaned reviews
- **Sentiment Class Distribution:**
 - Positive (Class 2): 56,107 reviews (~74.9%)
 - Neutral (Class 0): 16,286 reviews (~21.7%)
 - Negative (Class 1): 2,514 reviews (~3.4%)
- **Preprocessing Steps:**
 - Text cleaned by removing HTML tags, special characters, and extra spaces
 - Tokenization applied
 - Stopwords removed using NLTK
 - Lowercasing of text
 - Vectorization using both TF-IDF and Word2Vec methods

1. Evaluation Results

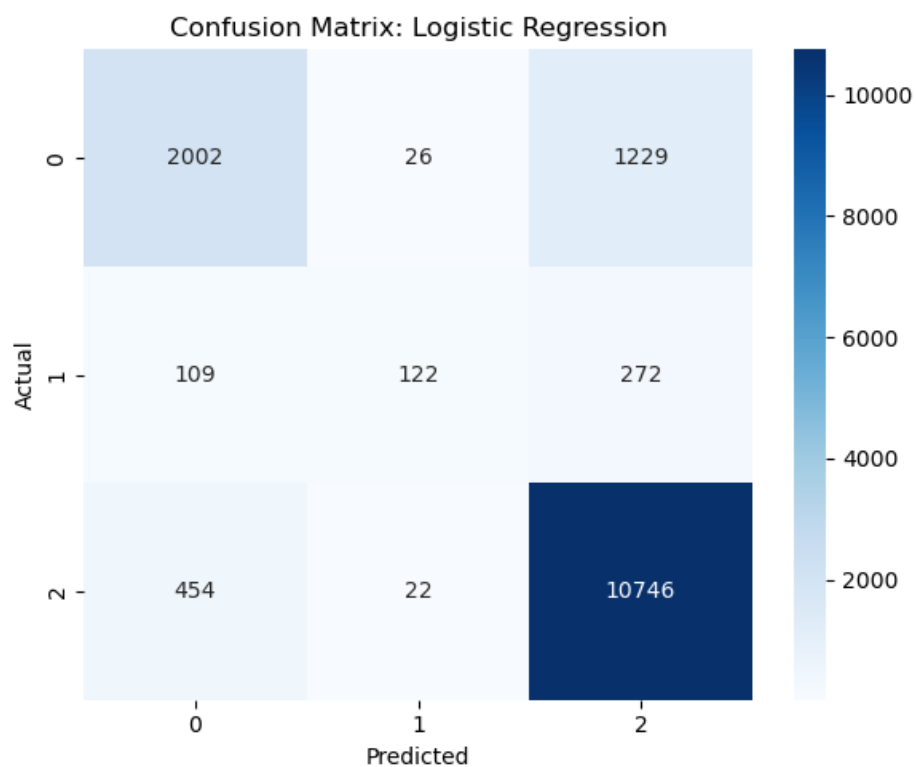
Bar Charts: Model Accuracy & F1-Score Comparison

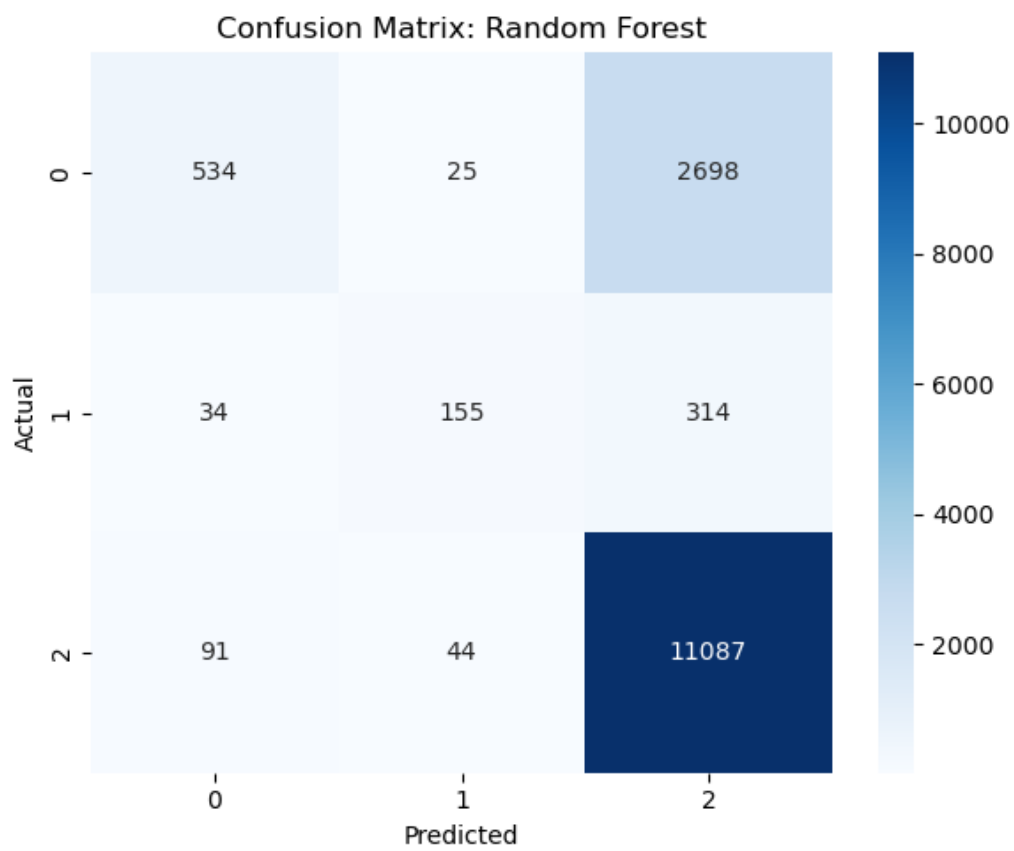
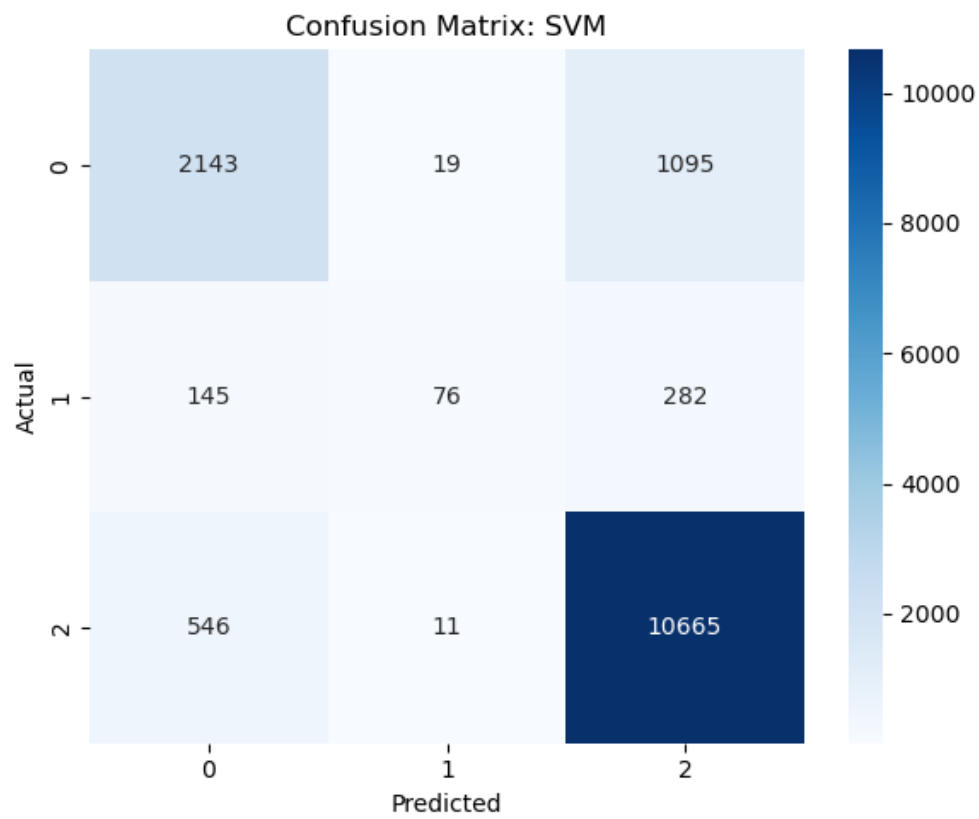
- **Accuracy Comparison**
 - Logistic Regression: 0.8590
 - SVM: 0.8600
 - Random Forest: 0.7860
- **F1 Score (Weighted) Comparison**
 - Logistic Regression: 0.8476
 - SVM: 0.8481
 - Random Forest: 0.7295



Confusion Matrices

- **Logistic Regression:** Shows high accuracy in classifying class 2 (positive sentiment), but lower recall for class 1 (negative sentiment).
- **SVM:** Slightly better than Logistic Regression across all classes, though still struggles with class 1.
- **Random Forest:** Performs well for class 2, but significantly underperforms for class 0 and class 1.





2. Challenges & Insights

- **Data Imbalance:**
 - Class 2 (Positive sentiments) dominates, leading to inflated accuracy and reduced recall for underrepresented classes.
- **Training & Computation Time:**
 - SVM required longer training time but showed the best generalization.
 - Random Forest was faster but underperformed.
- **Interpretability:**
 - Logistic Regression provides more explainability.
 - Feature importance could be extracted for Random Forest but was less interpretable for SVM.

Future modeling could benefit from balancing techniques and dimensionality reduction.

3. Updated Evaluation Report

- Performance visualizations added to the PDF:
 - Accuracy and F1-score bar charts
 - Confusion matrices for each model
- **Conclusions of Week 5:**
 - SVM is the best-performing model in current evaluations.
 - Logistic Regression remains a viable, interpretable alternative.
 - Random Forest may not be optimal without further tuning.
- **Improvement Areas:**
 - Apply SMOTE or other techniques for handling class imbalance.
 - Explore dimensionality reduction (e.g., PCA).
 - Experiment with ensemble methods or stacking.

4. Next Steps (Week 6 Roadmap)

Model Enhancements

- **Introduce Deep Learning Models:**
 - Implement **LSTM (Long Short-Term Memory)** networks to capture sequential patterns in text.
 - Fine-tune **BERT (Bidirectional Encoder Representations from Transformers)** using pretrained embeddings to improve semantic understanding.

Hyperparameter Tuning:

- Apply **Grid Search** or **Randomized Search** to optimize key hyperparameters of Logistic Regression, SVM, and Random Forest.
- Use **cross-validation** for robust performance estimation.

Deployment Preparation

- **Model Export & Inference:**
 - Convert best-performing models to deployable formats (.pkl, .joblib, .h5).
 - Develop REST API or Flask endpoints for serving predictions.
- **Documentation & Testing:**
 - Create model cards and validation reports.
 - Perform testing for model inference latency and accuracy in production-like settings.